

Impacting our community, region and world

Scale: More than **137,000 master learners**
across five campuses at the nation's largest
public research university.

Top 1% of world's
most prestigious
universities

#6
among U.S.
universities
for tech
company
hires

— SHL, 2020

#7
in the U.S.
for research
expenditures

ASU ahead of Princeton
and Caltech

National Science Foundation, 2019

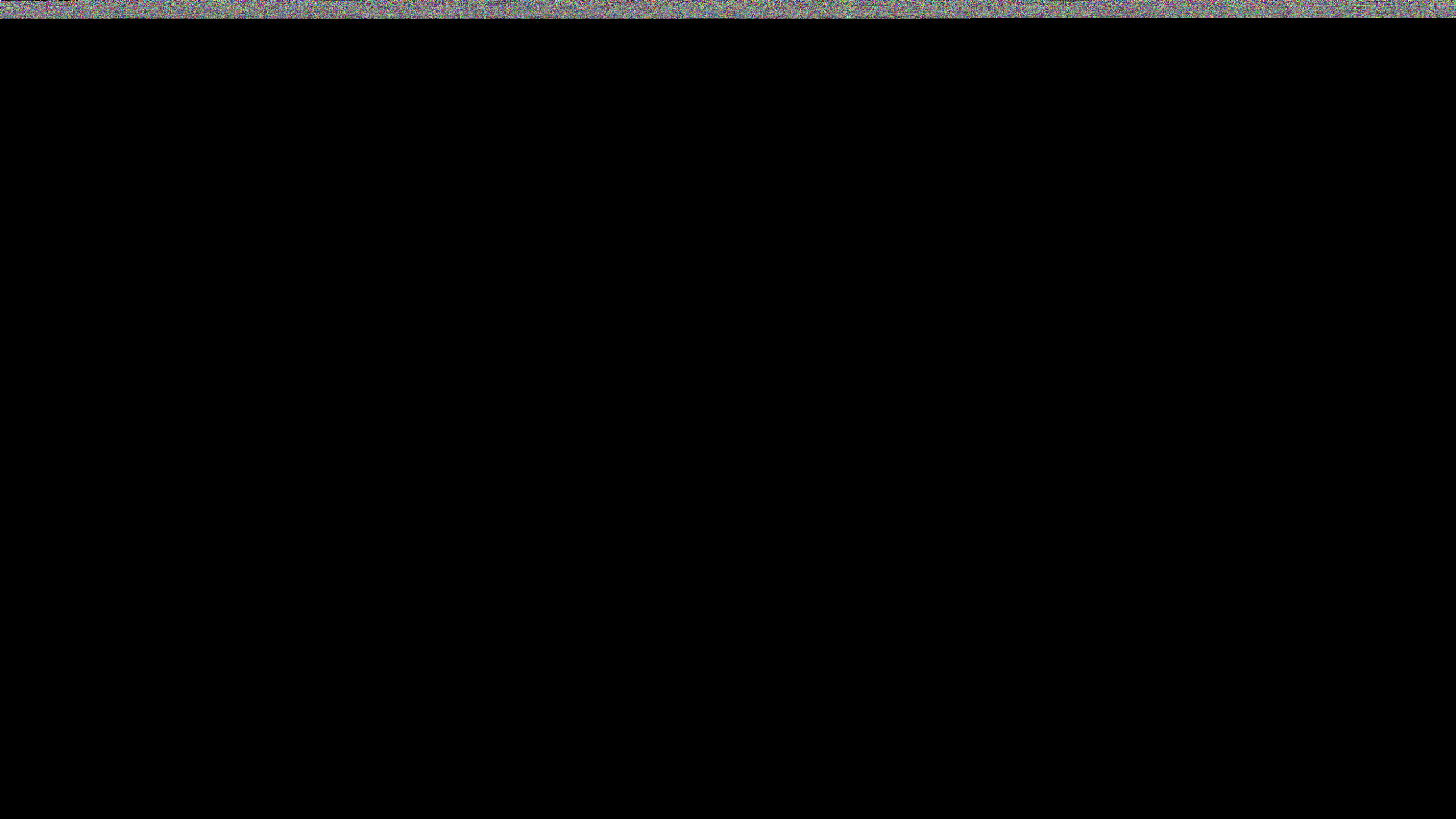
**Top
10**
in the nation
for patents

among universities
granted U.S. patents

U.S. National Academy
of Inventors and the
Intellectual Property
Owners Association

ASU Charter

ASU is a comprehensive public research university, measured not by whom it excludes, but by **whom it includes** and how they succeed; advancing **research and discovery of public value**; and assuming fundamental **responsibility for** the economic, social, cultural and overall health of **the communities it serves**.



FSE in Fall 2021

The largest and most comprehensive engineering college in the United States

Interdisciplinary structures

- **7 schools**; 2 campuses + online
- Cross-campus partnerships

Academic programs, enrollments and graduates

- 25 undergraduate degree programs; 50+ graduate programs
- **~27,000 students** (~9,000 online, ~18,000 on-campus)
- ~5,100 graduates across all degree levels in AY2020-2021

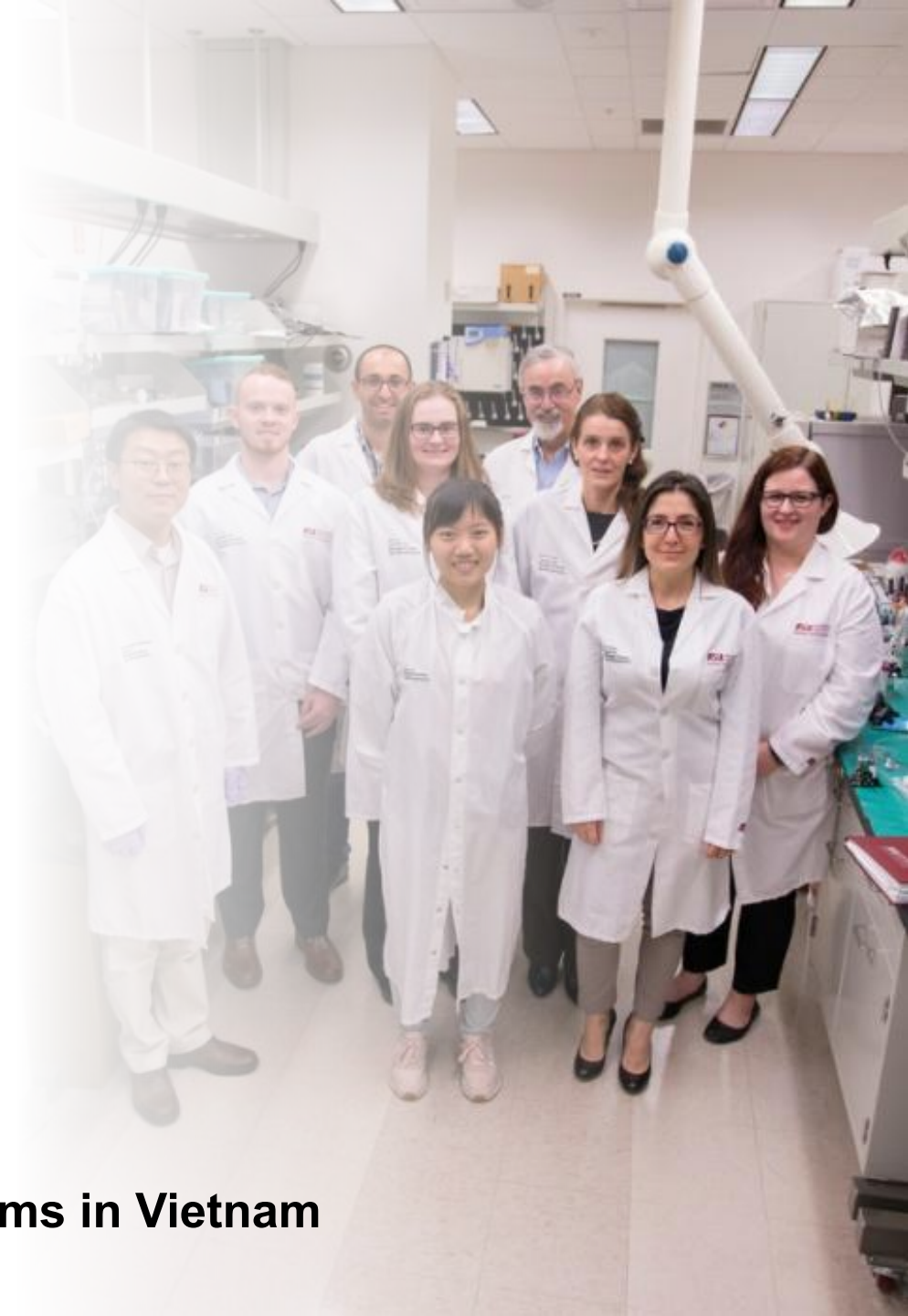
Faculty

- ~370 tenured/tenure-track faculty; ~100 lecturers and professors of practice
- **31 NSF CAREER awards** over the past three years

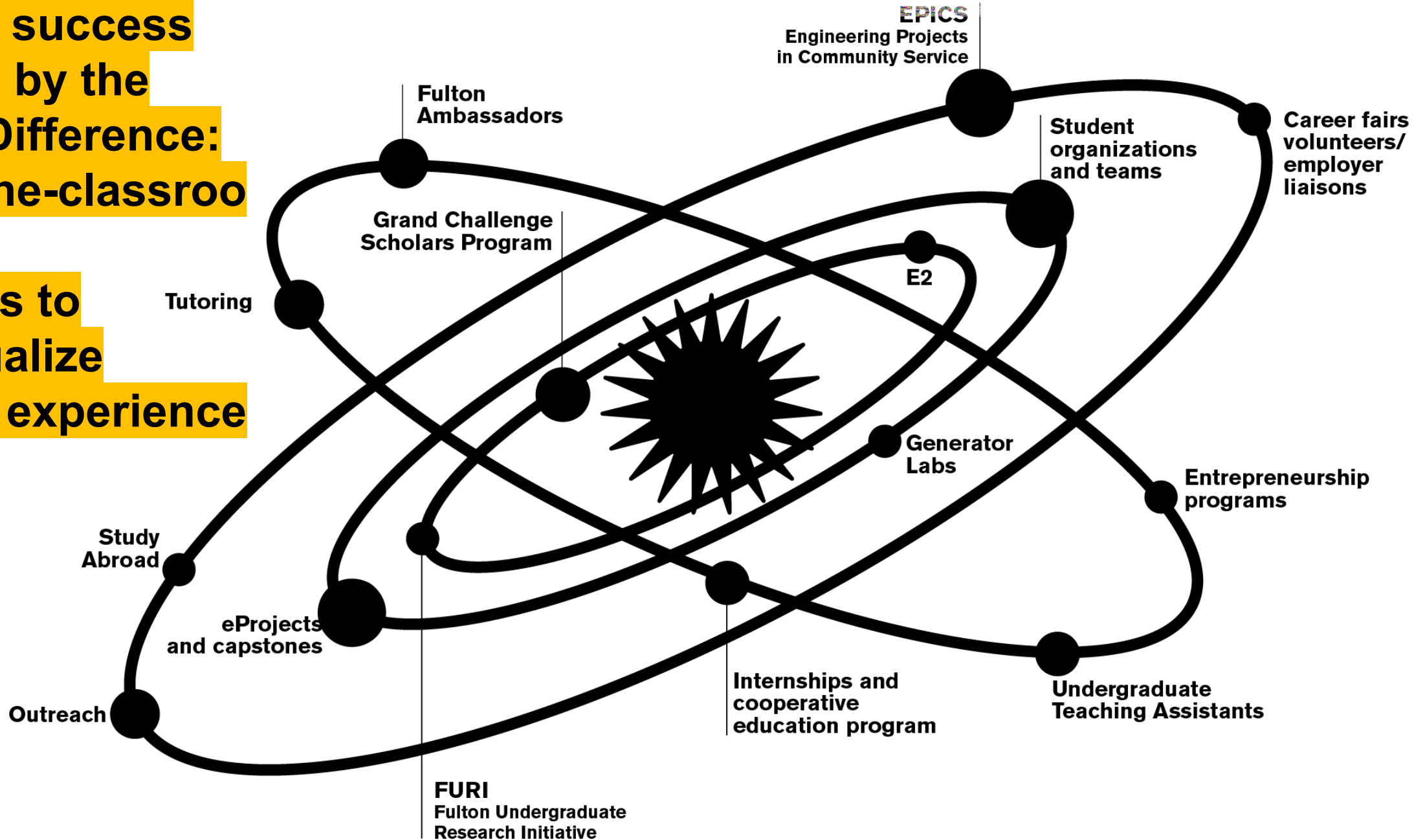
Research and innovation: \$141M in awards (10% increase)

- Lead two NSF ERC's (partner on one other ERC)
- Lead DHS Center for Accelerating Operational Efficiency
- Entrepreneurial outputs (national ranking per \$10M of research expenditures)
 - #7 for licenses & options; #6 IP disclosures; **#5 for startups**

Global connectivity: PLoS Alliance; capacity-building programs in Vietnam



**Student success
enabled by the
Fulton Difference:
out-of-the-classroom
activities to
individualize
student experience**



Advancing the Student Experience

FURI Fulton
Undergraduate
Research
Initiative
Fall 2020 Symposium

Join us at the **FURI Virtual Symposium**
Friday, November 20, 2020, 1–3 p.m.

furi.engineering.asu.edu



Environmental engineering major Emma Bonham works in the Keep Phoenix Beautiful Community Garden. Bonham worked with her mentor, Professor Rebecca Muenich, to assess urban farming's environmental sustainability.

**school of sustainable
engineering and the
built environment**

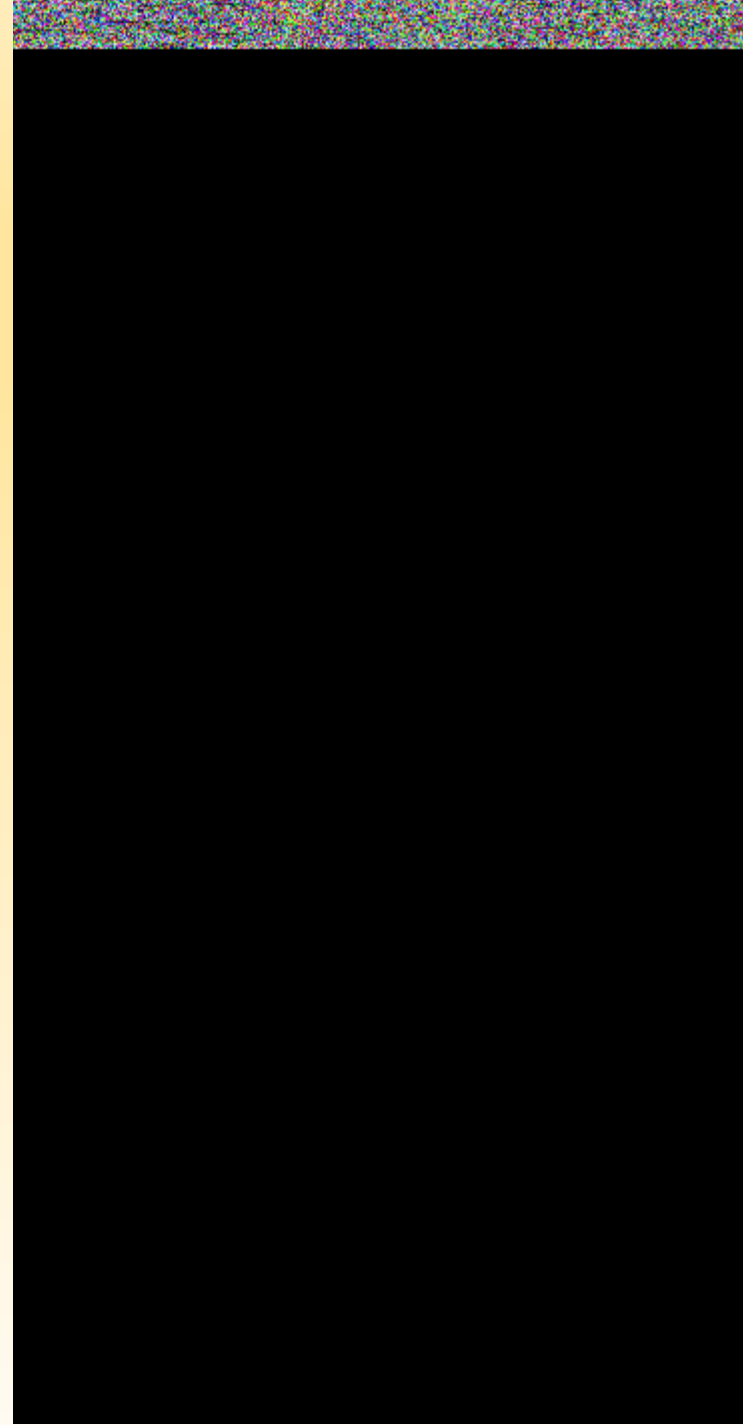
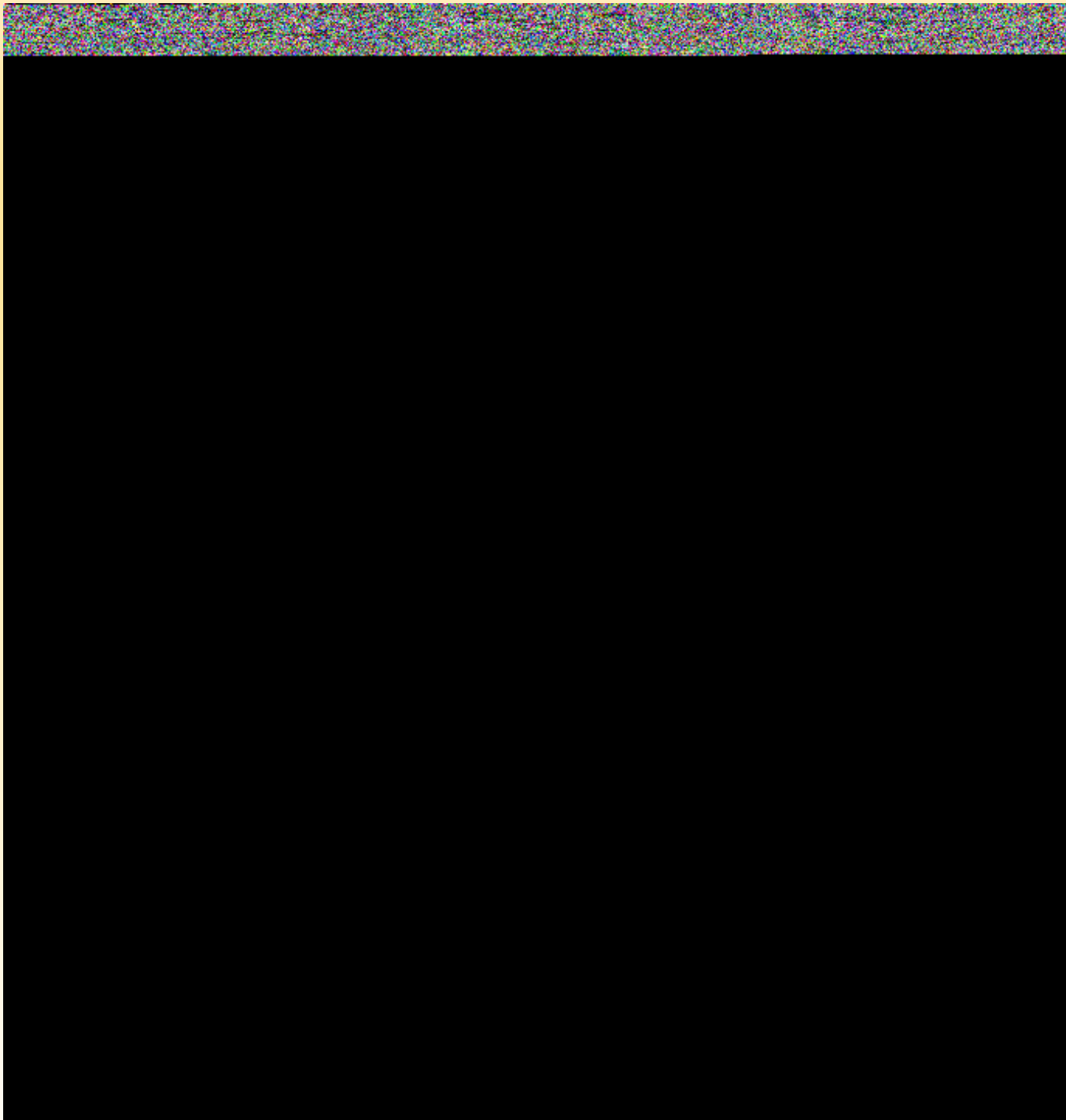


**del e. webb school of
construction**

civil, **environmental
and sustainable
engineering**

**construction
engineering**

**environmental
engineering**



SSEBE by the numbers

Fall 2021 Enrollment

 **187** environmental engineering

799 civil engineering 

 **108** construction engineering

721 construction management 



NSF Engineering Research Centers (SSEBE)

- Ed Kavazanjian, Director: **“NSF Engineering Research Center for *Bio-Mediated and Bio-Inspired Geotechnics*”**
 - \$36 million over ten years
 - Partners: ASU (Lead), UC Davis, New Mexico State Univ, Georgia Tech
 - <https://engineering.asu.edu/cbbg/>

- Paul Westerhoff: **“NSF Engineering Research Center for *Nanotechnology Enabled Water Treatment (NEWT)*”**
 - ~\$3.2 million over five years
 - Partners: Rice University (Lead), ASU, Yale, University of Texas El Paso
 - <http://www.newtcenter.org>

Examples of Centers in SSEBE - Variety of Engineering Research Activities

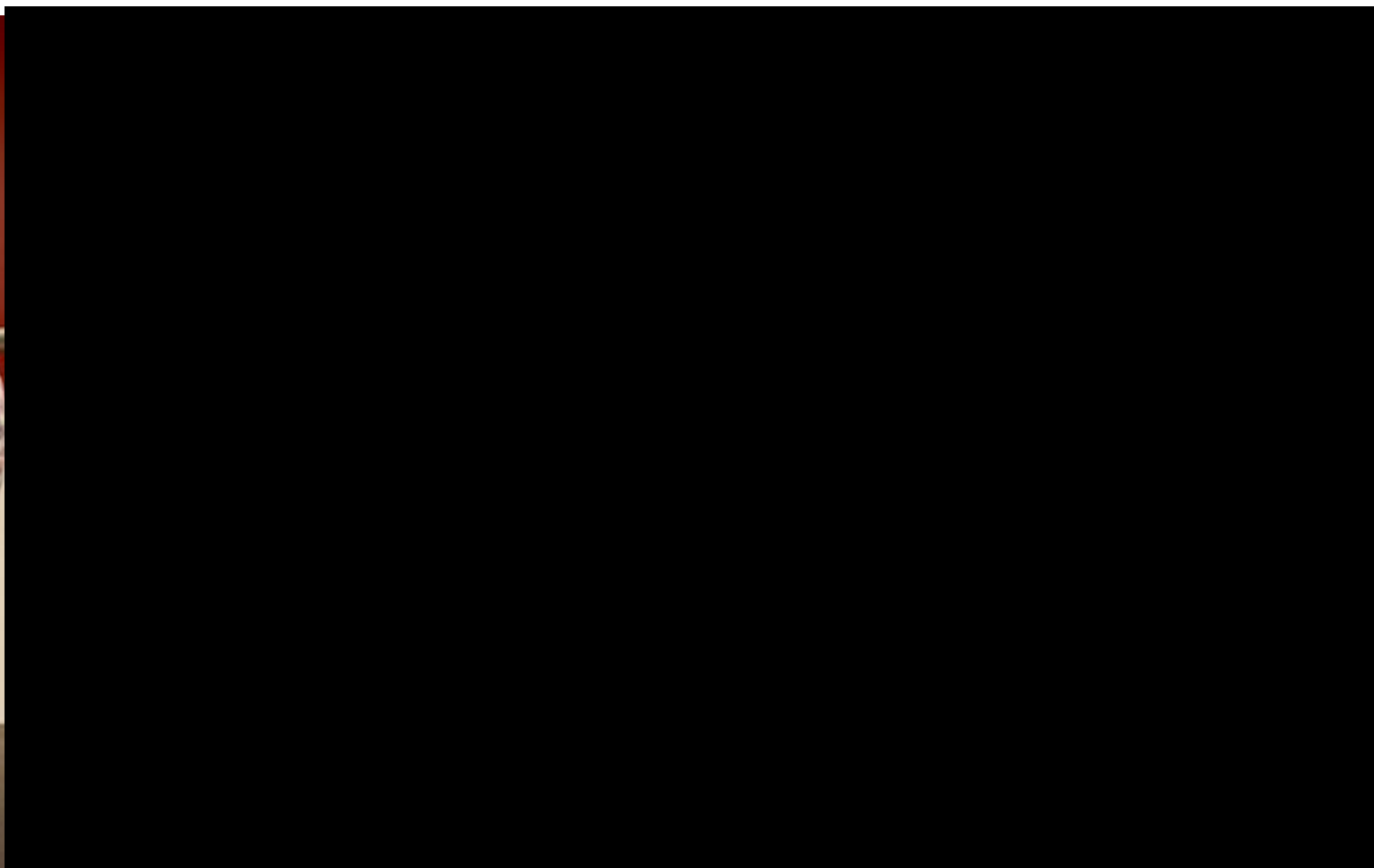
- Ram Pendyala, Director: **USDOT-sponsored Tier 1 University Transportation Center (TOMNET)**
 - Partners: ASU (Lead), Georgia Tech, Univ of Washington, Univ of South Florida
 - <https://www.tomnet-utc.org/>

- Kamil Kaloush □ Hasan Ozer, Director: **National Center of Excellence on SMART Materials for Urban Climate and Energy**
 - Impact: Technology, Materials, Business and Policy Innovations
 - <https://ncesmart.asu.edu/>

- Mikhail Chester, Director: **Metis Center for Infrastructure and Sustainable Engineering**
 - Impact: Resilience, Sustainability, Extreme Events, Climate Change Adaptation and Mitigation, Infrastructure as a Complex System
 - <https://metis.asu.edu/>

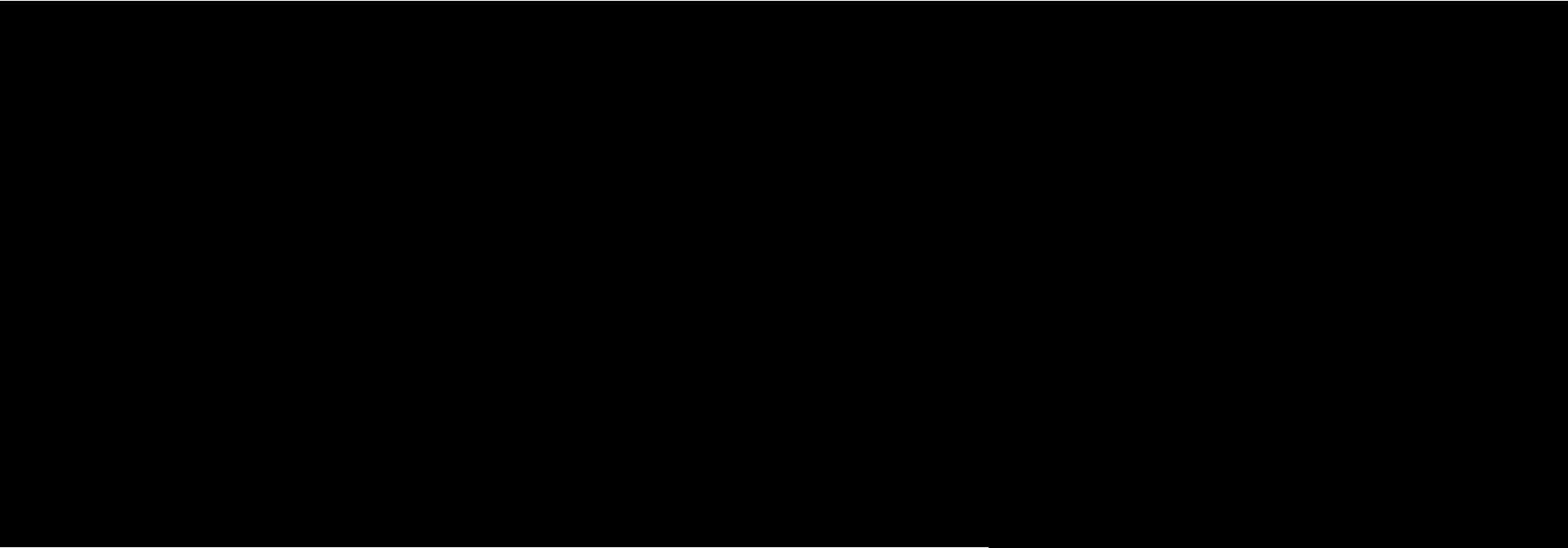
Center for Negative Carbon Emissions





ASU Engineering Professor Narayanan Neithalath and four colleagues have been granted \$2 million from the National Science Foundation to foster collaboration around 3D concrete printing research across more than a dozen countries. 3D concrete printing generated these examples shown with Sooraj Nair, a doctoral student in Neithalath's lab group, and the technique offers the potential to change the nature of construction.

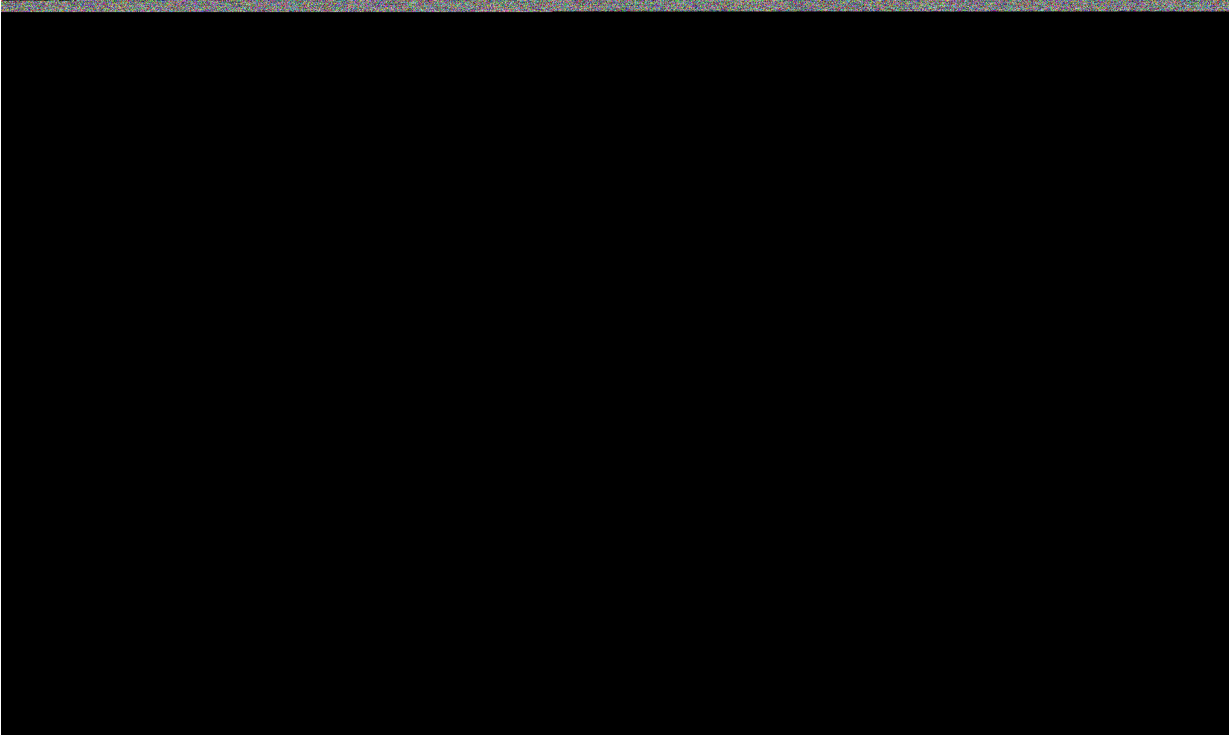
Photo courtesy of Narayanan Neithalath



Barzin Mobasher, professor in SSEBE of the Ira A. Fulton Schools of Engineering, is researching fiber-enforced concrete with the goal of saving time and money on construction projects. Photo by Charlie Leight/ASU Now

Award Winning Faculty

Edd Gibson Inducted as ASCE Distinguished Member Class of 2020



Professor George Edward “Edd” Gibson, Jr., considered the foremost expert in front-end planning of capital facilities in academia, is one of eight civil engineers nationwide to be named a Distinguished Member of the American Society of Civil Engineers in 2020.

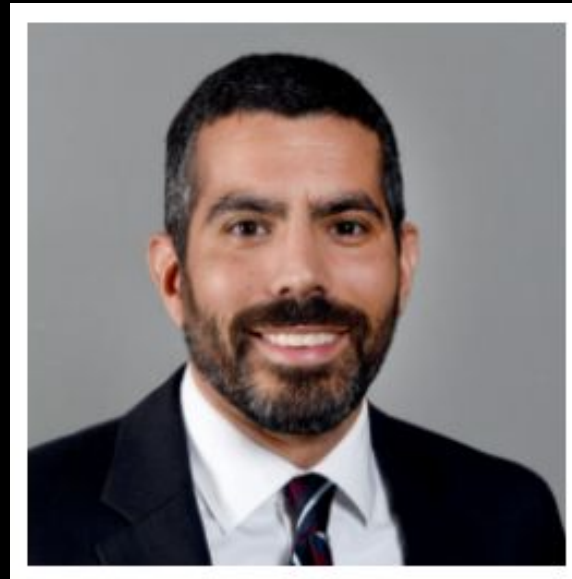
Kamil Kaloush Named FORTA Professor of Pavement Engineering



Kamil Kaloush, a professor in the Ira A. Fulton Schools of Engineering at ASU, will be able to broaden his academic and research endeavors with support that comes with his new position as the FORTA Professor of Pavement Engineering. Kaloush is pictured speaking at an IRF event.

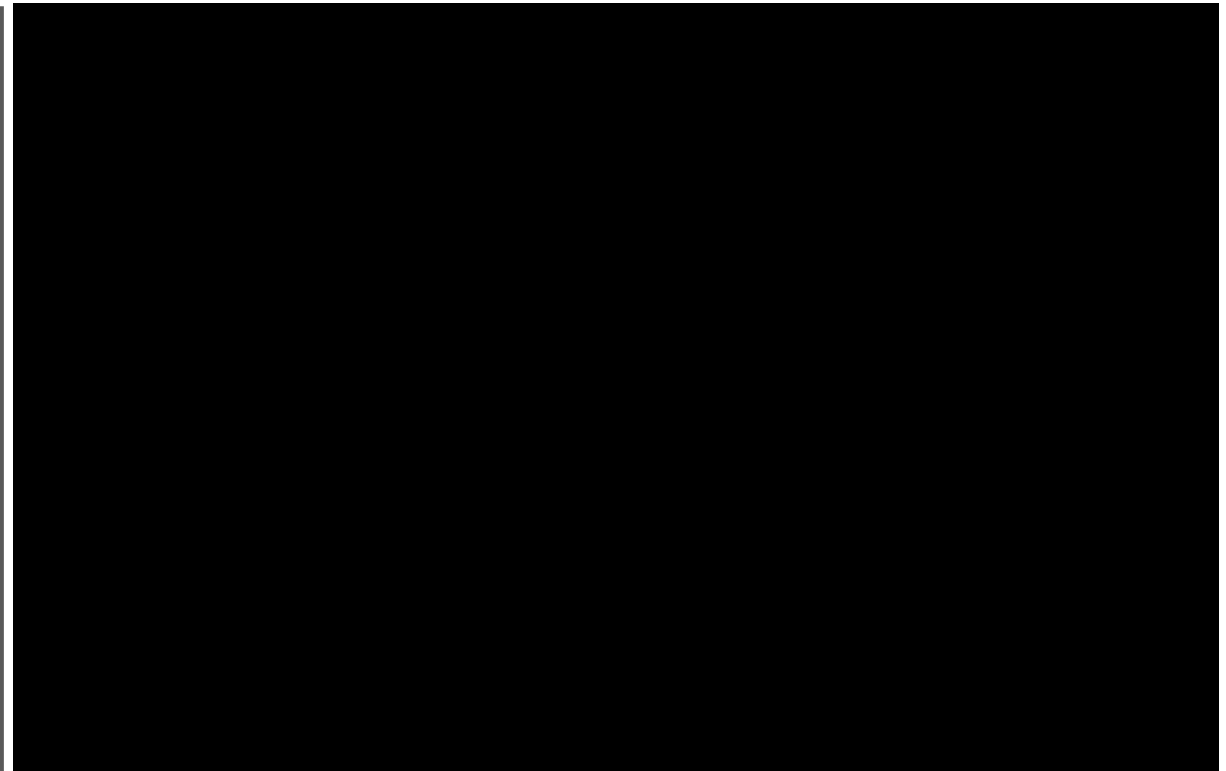
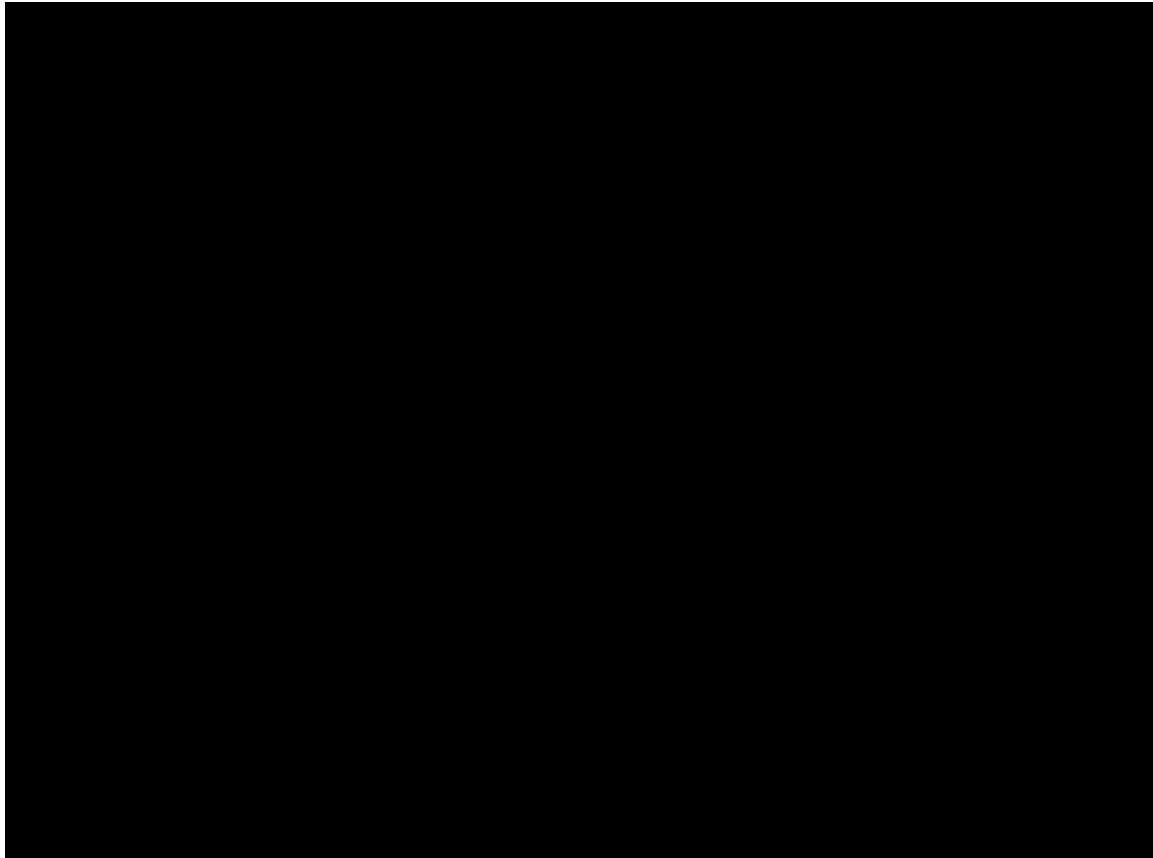
Washington Post

Why climate change is about to make your bad commute worse



ASU Transportation

Developing Partnerships and Solutions for A Better Transportation Future



Major Research Sponsors

- Arizona Department of Transportation
- Maricopa Association of Governments
- Maricopa County Department of Transportation
- Valley Metro
- National Cooperative Highway Research Program
- Transit Cooperative Research Program (TCRP)
- Airport Cooperative Research Program (ACRP)
- National Science Foundation
- USDOT, Federal Highway Administration/Federal Transit Administration
- USDOT, Research and Innovative Technology Administration
- USDOT, Intelligent Transportation Systems Joint Program Office
- National Science Foundation (NSF)
- US Department of Homeland Security (DHS)
- US Department of Energy (DOE)
- Southern California Association of Governments
- New York Metropolitan Transportation Council



TOMNET

Teaching **O**ld **M**odels **N**ew **T**ricks



US Department of Transportation
Tier 1 University Transportation Center

Advanced Data Analytics for
Sustainable Transportation

<http://www.tomnet-utc.org>

www.evsts.org

The Transportation Education Enterprise

Transportation courses and educational activities undertaken in multiple units

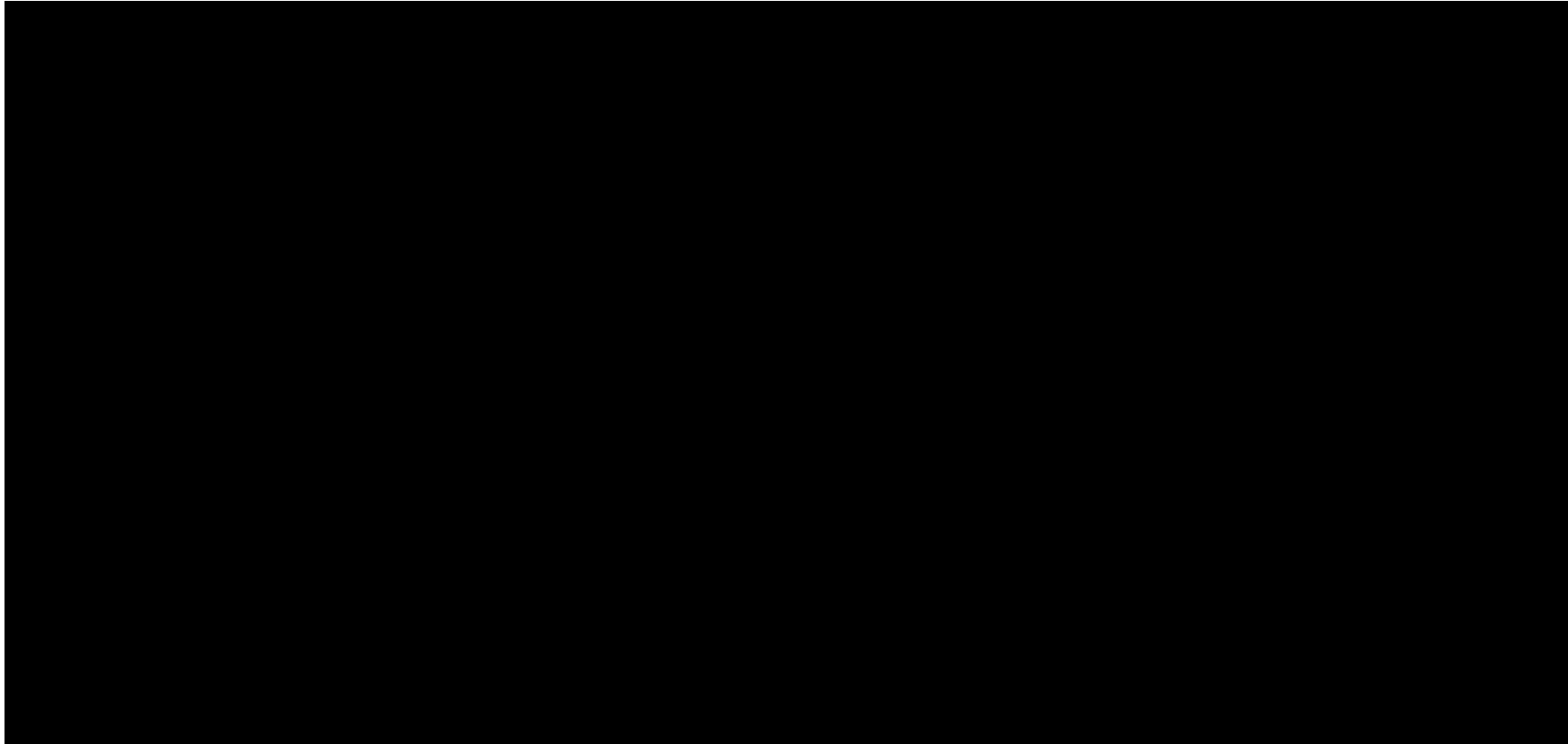
Courses in a multitude of domains

- Traffic engineering, operations, and control
- Transportation systems modeling and simulation
- Highway design, pavement design, and smart materials
- Transportation safety
- Transportation network optimization and modeling
- Public transportation planning and design
- Energy and environmental aspects of transportation
- Transportation planning and policy analysis
- Vehicular technologies, sensors, and autonomous systems

Strive to provide students multidisciplinary education experience

The Transportation Systems Certificate

- Graduate Certificate in Transportation Systems
- Students complete 15 credits of coursework across three different programs/schools and complete a capstone project
- Developing flexible **Online Master's Degree in Transportation**



Students Learn in the Real-World

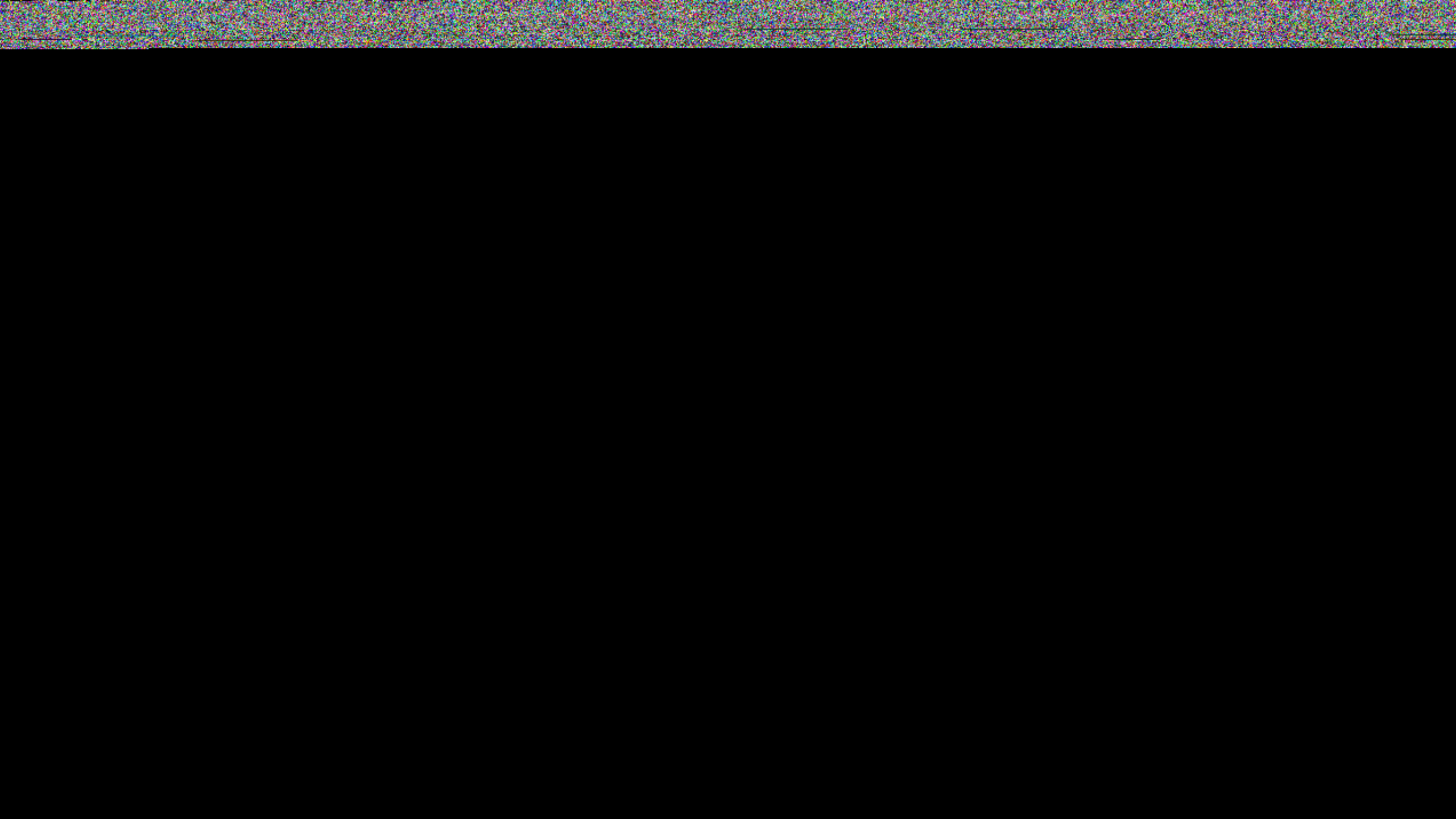
Pervious / Porous Parking Lots



The Future of Higher Education... Reimagined



<https://xr.asu.edu/>



Capstone - reimaged

- A **two-semester** capstone experience
- Full-time Professor of Practice dedicated to capstone, engaging industry deeply in the experience
- **Real world cross-cutting projects**
- Students work in cross-disciplinary teams
- Experiences essential to ensuring ***graduates are workforce ready***
- Projects bring together industry with their future workforce



school of **sustainable engineering** and the **built environment**

Ram M. Pendyala, ram.pendyala@asu.edu

For more information, please visit: <http://ssebe.engineering.asu.edu>

Thank you